Status	Finished
Started	Sunday, 23 August 2026, 4:27 PM
Completed	Thursday, 27 August 2026, 11:59 PM
Duration	4 days 7 hours
Marks	0.00/6.00
Grade	**0.00** out of 10.00 (**0**%)

Question 1

Not answered

Marked out of 1.00

Drag the pipeline stages into the correct execution order for a classic 5-stage RISC pipeline.

[ ] → [ ] → [ ] → [ ] → [ ]

EX (Execute)	MEM (Memory Access)	IF (Instruction Fetch)	ID (Instruction Decode)	WB (Write Back)

Your answer is incorrect.

The correct answer is:
Drag the pipeline stages into the correct execution order for a classic 5-stage RISC pipeline.

[IF (Instruction Fetch)] → [ID (Instruction Decode)] → [EX (Execute)] → [MEM (Memory Access)] → [WB (Write Back)]

Question 2

Not answered

Marked out of 1.00

Match each stage of the classic MIPS 5-stage pipeline to its primary function.

IF (Instruction Fetch)	Choose...
ID (Instruction Decode)	Choose...
EX (Execute)	Choose...
WB (Write Back)	Choose...
MEM (Memory Access)	Choose...

Your answer is incorrect.

The correct answer is: IF (Instruction Fetch) → Reads the instruction from instruction memory using the PC address, ID (Instruction Decode) → Reads register values and sign-extends the immediate field, EX (Execute) → ALU performs arithmetic/logic operations or calculates addresses, WB (Write Back) → Writes the final result back into the destination register, MEM (Memory Access) → Reads from or writes to data memory (for lw/sw instructions)

Question 3

Not answered

Marked out of 1.00

In a MIPS 5-stage pipeline, which pipeline register holds intermediate results between the Execute and Memory Access stages?

a. MEM/WB

b. EX/MEM

c. ID/EX

d. IF/ID

Your answer is incorrect.

The correct answer is: EX/MEM

Question 4

Not answered

Marked out of 1.00

Which of the following MIPS instructions is an I-type (Immediate-type) instruction?

a. `add $t0, $t1, $t2`

b. `addi $t0, $t1, 5`

c. `sub $s0, $s1, $s2`

d. `sll $t0, $t1, 2`

Your answer is incorrect.

I-type instructions include an immediate constant or offset. R-type instructions use only registers.

The correct answer is: `addi $t0, $t1, 5`

Question 5

Not answered

Marked out of 1.00

The ARM instruction `0xE0821003` has the following binary:

```
1110 0000 1000 0010 0001 0000 0000 0011
```

Drag the correct decoded values into each field:

- Condition code (bits 31–28): [ ]
- Operation / Opcode (bits 24–21): [ ]
- Destination register Rd (bits 15–12): [ ]
- First source Rn (bits 19–16): [ ]
- Second source Operand2 (bits 3–0): [ ]

R1	SUB (0010)	AL (Always execute)	R0	MOV (1101)	R3
ADD (0100)	R2	EQ (Equal)			

Your answer is incorrect.

The correct answer is:
The ARM instruction `0xE0821003` has the following binary:

```
1110 0000 1000 0010 0001 0000 0000 0011
```

Drag the correct decoded values into each field:

- Condition code (bits 31–28): [AL (Always execute)]
- Operation / Opcode (bits 24–21): [ADD (0100)]
- Destination register Rd (bits 15–12): [R1]
- First source Rn (bits 19–16): [R2]
- Second source Operand2 (bits 3–0): [R3]

Question 6

Not answered

Marked out of 1.00

An ARM data processing instruction has the hex value `0xE0421003`. What operation does this instruction perform?

Hint: Decode the condition code (bits 31–28), opcode (bits 24–21), Rn (bits 19–16), Rd (bits 15–12), and Operand2 (bits 3–0).

Recommended Steps:

1. Convert the hex values into a full binary string.
2. Map each bit to its corresponding decoded instruction part. *Watch-out for hex value splits across the bit fields!*
3. Lookup the ARM Instruction Set Reference guides *(images in MyCourses->Unit 3)*.

a. MOV R1, R2

b. SUB R2, R1, R3

c. ADD R1, R2, R3

d. SUB R1, R2, R3

Your answer is incorrect.

The correct answer is: SUB R1, R2, R3